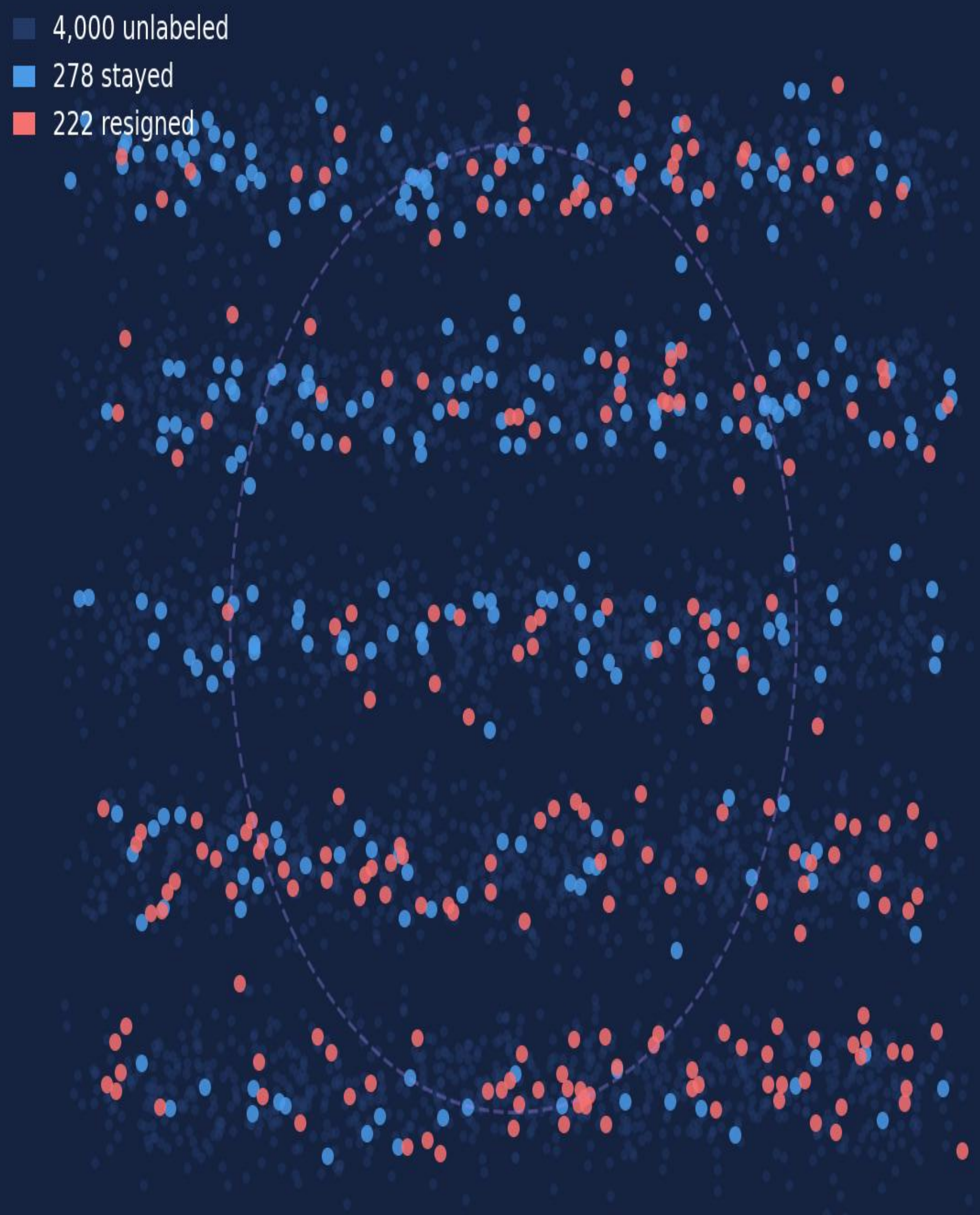
# Employee Turnover Prediction

Self-Training · Semi-Supervised

Project 13 · LozanoLsa Industrial ML Portfolio

*When 89.2% of your data has no label, you don't wait — you infer.*

Python | scikit-learn | RandomForest | Self-Training




**> 4,147**
Employee Profiles

**500**
Labeled Records

**0.705**
AUC-ROC Score

**332**
Pseudo-Labels

# The Real Problem



### Labels only accumulate through departures

A resignation record only exists after someone leaves. Every employee still working is, by definition, unlabeled. The dataset grows with time, but never catches up.

### Waiting is a strategy that always arrives too late

Organizations that delay modeling until they have 'enough' labeled data will always be predicting exits that already happened. The risk window has passed.

### 89.2% of what you know is being discarded

4,147 profiles contain real satisfaction scores, overtime patterns, and workload signals. A purely supervised approach throws all of that away because there is no label.

500 confirmed outcomes · 4,147 active unknowns · Self-Training bridges the gap

*Most organizations have far more observations than labels — and they keep collecting the wrong kind of data.*

# From Labeled Events to Probabilistic Inference



## ⊘ Old Thinking

— Collect resignation records.

— Train on labeled data only.

— Ignore 4,147 unlabeled profiles.

— Wait for more labels to accumulate.

— Model is always predicting the past.

— Coverage: 10.8% of the workforce.

## ✓ New Thinking

✓ Start with 400 labeled training records.

✓ Score 4,147 unlabeled profiles.

✓ Promote only high-confidence predictions.

✓ Threshold: $P \geq 0.90$ or $P \leq 0.10$.

✓ Expand to 732 training examples.

✓ Evaluate on real labels only. Always.

*Self-Training does not invent labels. It promotes only those predictions the base model is already most confident about.*

# The Data Behind the Problem



| | | |
|---|---|---|
| **HRIS**<br>SAP / Workday | **age · years_at_company · job_level** | Core employee record — available from day one of hiring. |
| **Payroll**<br>ADP / SAP HR | **monthly_salary** | Exact gross compensation — updated each payroll cycle. |
| **Survey**<br>Glint / Qualtrics | **job_satisfaction · workload** | Quarterly pulse surveys — the voice of the employee in the data. |
| **T&A**<br>Time & Attendance | **avg_overtime_hours** | Monthly aggregation of clock records — proxy for workload-contract mismatch. |

## What makes this dataset unique

- 7 features — zero preprocessing needed. Random Forest is scale-invariant.
- 500 labeled records (10.8%) — rare enough to matter, enough to bootstrap.
- 4,147 unlabeled rows are not noise. They are signals waiting to be read.
- Satisfied high-salary employees rarely resign — the model learned this cleanly.
- Low satisfaction + high workload = near-100% resignation rate in labeled set.

*The label column (resigned) is the rarest variable. Everything else is collected in real-time by operational systems.*

# Why Self-Training?



*"Semi-supervised learning exists because labels are expensive and observations are cheap. Self-Training is the simplest honest answer to that asymmetry."*

### 01 Label efficiency

Supervised training uses 400 examples. Self-Training uses 732. That 83.0% expansion comes from data we already had — salary, satisfaction, overtime — just without a confirmed outcome attached.

### 02 Confidence-gated expansion

Only predictions at $P \geq 0.90$ or $P \leq 0.10$ enter training. 3,815 employees in the uncertain middle are deliberately excluded. Selectivity protects signal quality.

### 03 No preprocessing pipeline

Random Forest is scale-invariant and handles mixed numeric types natively. No normalization, no encoding. The model earns its keep by reducing infrastructure complexity.

### 04 Evaluation stays honest

The 100-record test set was separated before any pseudo-labeling occurred. Metrics reflect performance on real labels only — not the model's own confidence.

*The algorithm does not invent knowledge. It extends what the model already believes with high confidence.*

# Model Results



| 69.0% | 0.705 | 0.608 | 68.6% |
|---|---|---|---|
| Accuracy | AUC-ROC | F1 Score | Precision |

### Top Drivers — Feature Importance

| | |
|---|---|
| job_satisfaction | 0.262 |
| monthly_salary | 0.151 |
| workload | 0.184 |
| avg_overtime_hours | 0.161 |
| years_at_company | 0.100 |
| age | 0.101 |
| job_level | 0.040 |

### Confusion Matrix  (n = 100 test records)

| | Pred: Stay | Pred: Resign |
|---|---|---|
| Actual: Stay | 45 ✓ | 11 ✗ |
| Actual: Resign | 20 ✗ | **24 ✓** |

*Recall: 54.5%  —  Of 44 real resigners, the model flagged 24*

*Evaluated on 100 real held-out records. Pseudo-labels never entered the test set. AUC-ROC 0.705 — meaningful ranking with label-scarce training data.*

# From Insight to Action



job_satisfaction 0.262

**Satisfaction Signal**

→ Quarterly pulse checks for Level 1–2 employees. Flag anyone scoring ≤ 2 for immediate 1:1 with their manager. Do not wait for the annual survey.

monthly_salary 0.151

**Compensation Anchor**

→ Annual market benchmarking by role and level. Proactive adjustments for employees in the 6–10 year tenure band before they begin job searching.

workload 0.184

**Workload Redistribution**

→ Trigger HR review when workload self-score ≥ 4 persists for 60+ days. Overloaded employees rarely self-report — the data reports for them.

avg_overtime_hours 0.161

**Overtime Cap**

→ Structured policy: HR review activated when average overtime exceeds 15 h/month for two consecutive months. Chronic overtime is a contract violation in slow motion.

years_at_company 0.100

**Tenure Milestone Review**

→ Career path review at Years 3, 5, and 8. These are the departure risk windows where unspoken ambitions collide with perceived ceilings.

*Prediction without intervention is just expensive reporting. Each driver maps to a specific retention lever.*

# What-If Simulator



### Scenario A · Burnout Operative

| | |
|---|---|
| Age | 28 |
| Tenure | 6 yrs |
| Level | 1 — Operative |
| Salary | $9,000 |
| Satisfaction | 1 / 5 |
| Workload | 5 / 5 |
| Overtime | 18 h/mo |

**P(resign) = 91%**

### Scenario B · Stable Analyst

| | |
|---|---|
| Age | 33 |
| Tenure | 3 yrs |
| Level | 2 — Analyst |
| Salary | $21,000 |
| Satisfaction | 4 / 5 |
| Workload | 3 / 5 |
| Overtime | 4 h/mo |

**P(resign) = 1%**

### Scenario C · Manager / High Load

| | |
|---|---|
| Age | 42 |
| Tenure | 8 yrs |
| Level | 4 — Manager |
| Salary | $60,000 |
| Satisfaction | 3 / 5 |
| Workload | 5 / 5 |
| Overtime | 15 h/mo |

**P(resign) = 32%**

*Scenario C insight: A $60,000 salary does not neutralize satisfaction=3 + workload=5 + 15h overtime. Risk sits at 31.5% — a clear HR intervention signal.*

*Scenario C shows that even a high salary does not offset low satisfaction and sustained overtime for a manager-level employee.*

# Learning from what you don't yet know.

*Self-Training asked the base model one question: what do you already believe, and how confident are you?*
*Only the most certain answers made it into training. The rest stayed uncertain — and that restraint is what made the model trustworthy.*

AUC-ROC 0.705 on a label-scarce HR problem. Not magic — epistemology.

Full dataset (4,647 rows) · app.py simulator · PPTX deck
*Challenge the threshold. Set it to 0.80. Does the model improve or collapse?*
**Because the boundary between labeled and unlabeled has no finish line.**